

Levels of Measurement Clarified

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Levels of Measurement

RATIO

RATIO – exact measurement on a scale that has a true zero; e.g. height in cm, weight in pounds, drug dose in mg

INTERVAL

INTERVAL – discrete units on a scale, usually whole numbers; e.g. number of children, age decade (20-30, 30-40, 40-50, etc).

ORDINAL

ORDINAL – ordered categories or levels, but no underlying reference scale; e.g. early vs late; low vs moderate vs high

NOMINAL

NOMINAL – Categories, no order specified or implied; e.g. marital status, gender, hospital unit, blood type



Lower Precision
Less Information

Points of confusion

- Nominal versus Ordinal – if you specify an order to the categories, then you can analyze the data as ordinal
- For example, consider creating categories of Urban, Suburban or Rural to addresses by Zipcode.
- If you want to consider these location categories as proxies for access to healthcare resources where rural has the least < with suburban having more and < urban having the most access, then you can consider location category as Ordinal.
- **Depends on how the variable will be operationalized/conceptualized.**

Points of confusion

- Ordinal versus Interval – the crux of this distinction is whether the ranking of the categories or levels are evenly spaced (as defined on a reference scale). If evenly spaced, the variable is interval, if no reference can be provided or the category levels are not evenly spaced or you can't tell, then the variable is ordinal.
- Example – number of blocks walked. If we assume or have defined that each “block” is 50 feet long and all blocks are the same, then number of blocks walked is an interval variable. If the blocks vary in distance or we don't know if they vary, then number of blocks walked is an ordinal variable.

Tricky Question? Pain Rating

- “Rate your pain today from 0-10”
- What kind of variable is this?
 - Categorical?
 - Ordinal?
 - Interval?
 - Ratio?

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- “Rate your pain today from 0-10, none to worse ever”
- What kind of variable is this?
 - Categorical? – no there is a definite order
 - Ordinal? –
 - Interval? – the implied scale is interval since we have even whole numbers 0, 1, 2, ..., 9, 10. But does the pain increase by the same amount if someone changes from a 1 to 2 versus a change from 9 to 10?
 - Ratio? – even though there is a “true” zero, but probably not since whole numbers most likely reported, doubtful someone says my pain is 3.2 today.

Tricky Question? Pain Rating

- “Rate your pain today from 0-10, none to worse ever”
- What kind of variable is this?
 - Categorical? – no there is a definite order
 - **Ordinal? – yes, there is an order**
 - Interval? – the implied scale is interval since we have even whole numbers 0, 1, 2, ..., 9, 10. But does the pain increase by the same amount if someone changes from a 1 to 2 versus a change from 9 to 10? Is everyone’s “scale” the same? no
 - Ratio? – even though there is a “true” zero, but probably not since whole numbers most likely reported, doubtful someone says my pain is 3.2 today.

Points of confusion

- Interval versus Ratio – the key to distinguishing these 2 levels of measurement is mainly on how precise the measurement is and whether the variable has a “true” zero (more on this in a bit).
- Typically, interval variables are whole discrete numbers (integers) that cannot be (or are traditionally not) reported with fractions. For example, number of children – 3 children (not 2.75 children).
- Ratio variables often are (or can be) measured to a fractional part of the scale used to make the measurement. For example, height in inches: subject 5 is 65.2 inches tall.

Tricky Question? Age

- “How old are you?”
- This depends on how the data is captured and measured.
- Ratio: Age to the fractional year: 55.12 years old
- Interval: Age level (often provided as a choice within 5 or 10 year intervals to help keep data anonymous): >55 to 60 years old
- Ordinal: Age generation: “Generation X” [born 1965-1979]

<https://www.careerplanner.com/Career-Articles/Generations.cfm>

Tricky Question? Time

	Baseline Date	Follow-up 1 Date	Follow-up 2 Date	Follow-up 3 Date
Actual Date	03/01/2019	05/05/2019	09/01/2019	12/31/2019
Study time point	1	2	3	4
Days elapsed	0	65	184	305

- Study time point - is technically ordinal since the number of days between each sequential time point is a different number of days
- Days elapsed – could be treated as interval whole days, but can also be treated as a continuous ratio variable, since we could compute fractional days if we had exact times to go with each date listed

Very Tricky? Temperature

- Temperature is tricky since degrees Fahrenheit and Celsius do not have a “true” zero (where no temperature exists) – so technically these temperature scales are not ratio variables.
- Temperature in Kelvin is ratio since 0 Kelvin is the complete absence of temperature. This really only matters to physicists.
- For all intensive purposes, everyone analyzes temperature as a continuous (not as an interval) variables (like ratio variables).

Why does (any of) this matter?

Statistics that are appropriate	Nominal	Ordinal	Interval	Ratio
Mode, Counts, Frequencies, Percentages	☒	☒	☒	☒
Median, Minimum, Maximum, Range, Percentiles	☐	☒	☒	☒
Mean, Variance, Standard Deviation	☐	☐	☒	☒

Often Abused – LiKert Scales

- You will often see LiKert scale questions used in surveys
- For example:
 - 1 = Strongly Disagree
 - 2 = Disagree
 - 3 = Neutral
 - 4 = Agree
 - 5 = Strongly Agree
- You will see people report means and standard deviations for these data, but they are Ordinal data only
- Is the “change” from a 1 to 2 the same as from 3 to 4? Probably not
- You should report Medians and Percentiles for these type data